UNIVERSITY OF AGRICULTURE

Department of Computer Science

Lab 1: Programming Fundamentals

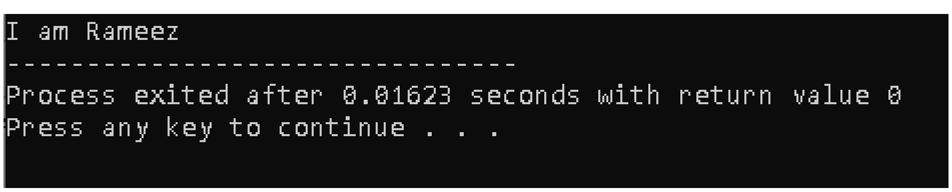
(CS-308)

Basic Information			
Lab #:	1	**Teacher:**	Arsal Mahmood
Objectives	1. Basic programming logics 2. Writing simple programs 3. Variables, datatypes and their uses		
Registration#	2025-ag-9987	**Student Name**	Syed Rameez Haider Rizvi
Total Marks	10	**Marks Obtained**	
Tools	Dev C++ / Visual Studio		

Activity 1 : Variables and Datatypes

1. Write a simple program to print your name on screen.

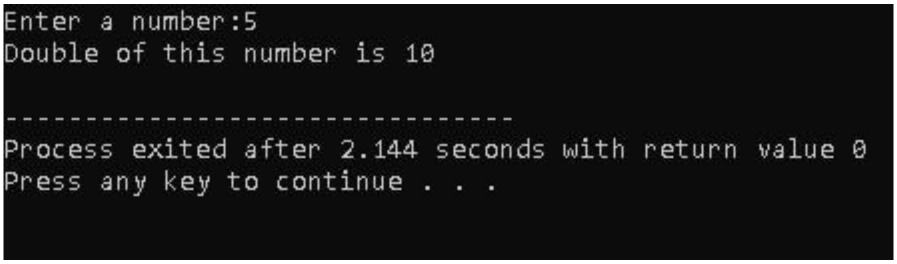
```
    #include <iostream>
using namespace std;        int
main(){
    cout<<"I am Rameez";
    }
```



2. Write a program to input a number from the user, double it and display it on the screen.

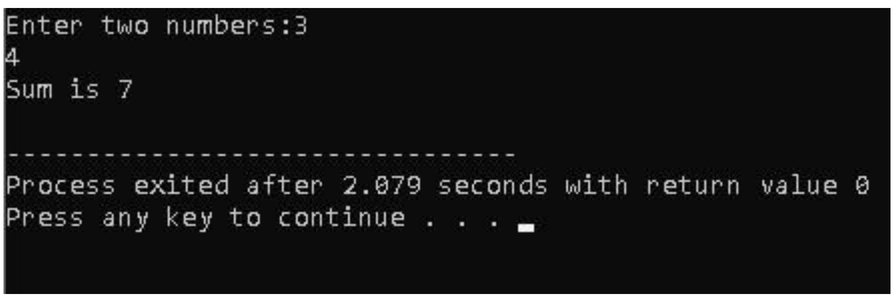
```
#include <iostream>
using namespace std; int
main(){
    int n;
    cout<<"Enter a number:";
    cin>>n;
```

```
    cout<<"Double of this number is "<<n+n<<endl;
}
```



3. Write a program to input two numbers and display their sum.

```
#include <iostream>
using namespace std;
int main(){  int a,b;
 cout<<"Enter two numbers:";
cin>>a>>b;
 cout<<"Sum is "<<a+b<<endl;
}
```



4. Write a program to calculate the area of a rectangle by taking length and width as input from the user.

```
#include <iostream>
using namespace std;
int main(){ float a,l,w;
cout<<"Enter Length:";
cin>>l;
cout<<"Enter Width:";
cin>>w; a=l*w;
cout<<"Area is "<<a;
}
```

5. Get user given values in variables a,b and c and display answer for the following equation:

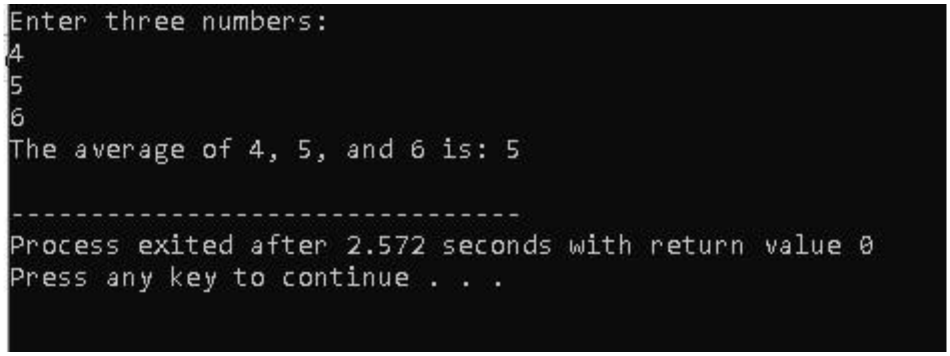
$$8ax^2 + 6bx + 4c$$

```
#include <iostream> using namespace std;

int main() {     float a, b, c, x, result;     cout

<< "Enter value for a: ";     cin >> a;     cout

<< "Enter value for b: ";     cin >> b;     cout

<< "Enter value for c: ";     cin >> c;     cout <<

"Enter value for x: ";     cin >> x;     result = (8

* a * x * x) + (6 * b * x) + (4 * c);     cout <<

"The result is: " << result << endl;

}
```

```
Enter value for a: 5
Enter value for b: 4
Enter value for c: 3
Enter value for x: 2
The result is: 220

--------------------------------
Process exited after 3.056 seconds with return value 0
Press any key to continue . . .
```

6. Write a program to calculate the average of three numbers entered by the user.

```
#include <iostream>

using namespace std; int

main() {

   float num1, num2, num3, average;
   cout << "Enter three numbers:" << endl;

cin >> num1 >> num2 >> num3;     average

= (num1 + num2 + num3) / 3;

   cout << "The average of " << num1 << ", " << num2 << ", and " << num3 << " is: " << average << endl;

}
```


```
Enter three numbers:
4
5
6
The average of 4, 5, and 6 is: 5

--------------------------------
Process exited after 2.572 seconds with return value 0
Press any key to continue . . .
```

7. Write a program to input the radius of a circle and calculate its area. #include

<iostream> using namespace std; int main() { float radius, area;

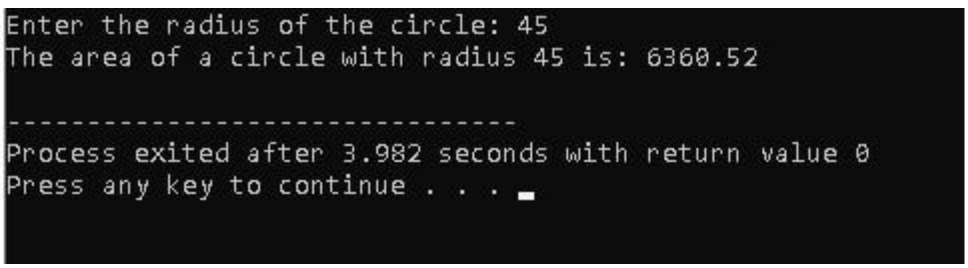
cout << "Enter the radius of the circle: ";

cin >> radius; area =

3.141*radius*radius;

cout << "The area of a circle with radius " << radius << " is: " << area << endl;

}


```
Enter the radius of the circle: 45
The area of a circle with radius 45 is: 6360.52

--------------------------------
Process exited after 3.982 seconds with return value 0
Press any key to continue . . .
```

8. Write a program to convert temperature from Celsius to Fahrenheit. #include

<iostream> using namespace std; int main() { float c, f;

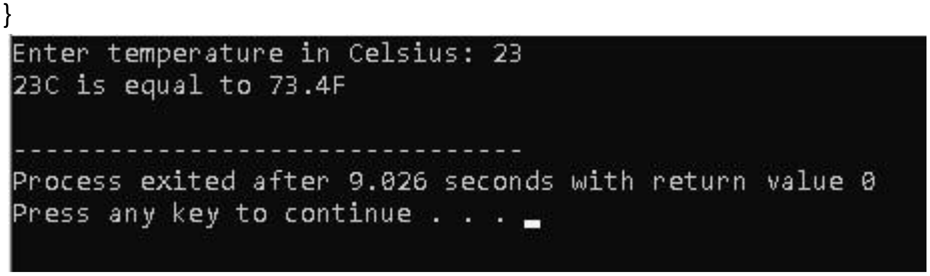
cout << "Enter temperature in Celsius: ";

cin >> c;

f = (c * 9.0 / 5.0) + 32;

cout << c << "C is equal to " << f << "F" << endl;

}


```
Enter temperature in Celsius: 23
23C is equal to 73.4F

--------------------------------
Process exited after 9.026 seconds with return value 0
Press any key to continue . . .
```

9. Input username, age, gender and city and store them in relevant variables. Then display on screen like: Hello **"name",** You're a **"age"** years old **"gender"** living in **"city"**.

#include <iostream>

using namespace std; int

main() {

string name, gender, city;

int age;

```
   cout << "Enter username: ";
cin>>name;     cout << "Enter
age: ";     cin >> age;
   cout << "Enter gender: ";
cin >> gender;      cout <<
"Enter city: ";     cin >> city;
  cout<<"Hello "<<name<< " You're a "<<age<< " years old "<<gender<< " living in "<<city<<endl;
}
```

10. Develop a calculator which can handle two user given numbers. The calculator can perform addition, subtraction, multiplication and division and display all the results on screen.

```
#include <iostream> using
namespace std; int main() {
float num1, num2;     cout <<
"Enter first number: ";     cin >>
num1;
   cout << "Enter second number: ";
cin >> num2;
   cout << "Addition:       " << num1 << " + " << num2 << " = " << (num1 + num2) << endl;
cout << "Subtraction:    " << num1 << " - " << num2 << " = " << (num1 - num2) << endl;
cout << "Multiplication: " << num1 << " * " << num2 << " = " << (num1 * num2) << endl;     if
(num2 != 0) {
      cout << "Division:       " << num1 << " / " << num2 << " = " << (num1 / num2) << endl;
   } else {
      cout << "Division:       Error! Cannot divide by zero." << endl;
   }
}
```

```
Enter first number: 5
Enter second number: 6
Addition:       5 + 6 = 11
Subtraction:    5 - 6 = -1
Multiplication: 5 * 6 = 30
Division:       5 / 6 = 0.833333

--------------------------------
Process exited after 6.109 seconds with return value 0
Press any key to continue . . .
```

11. Write a program that takes principal amount, rate of interest, and time from the user and calculates the simple interest using the formula:

$$SI = \frac{P \times R \times T}{100}$$

The program should display the calculated simple interest.

```
#include <iostream>
using namespace std;
int main() {
  float principal, rate, time, simple_interest;
cout << "Enter the Principal amount: ";
cin >> principal;
  cout << "Enter the Rate of interest (in %): ";
cin >> rate;
  cout << "Enter the Time (in years): ";
cin >> time;
  simple_interest = (principal * rate * time) / 100.0;
cout << "Principal: " << principal << endl;    cout <<
"Rate:     " << rate << "%" << endl;
  cout << "Time:     " << time << " years" << endl;    cout <<
"Simple Interest (SI) = " << simple_interest << endl;
}
```

```
Enter the Principal amount: 1000
Enter the Rate of interest (in %): 5
Enter the Time (in years): 2
Principal: 1000
Rate:      5%
Time:      2 years
Simple Interest (SI) = 100

--------------------------------
Process exited after 8.111 seconds with return value 0
Press any key to continue . . .
```

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12. Write a program that asks the user to enter the length of one side of a square. The program should then calculate: Display both the area and perimeter.

$$\text{Perimeter} = 4 \times \text{side}$$

$$\text{Area} = \text{side} \times \text{side}$$

```
#include <iostream>
using namespace std;
int main() {
  float side, area, perimeter;
cout << "Enter the length: ";
cin >> side;    perimeter = 4 *
side;    area = side * side;
  cout << "Perimeter:   " << perimeter << endl;
cout << "Area:        " << area << endl;
}
```

```
Enter the length: 12
Perimeter:    48
Area:         144

--------------------------------
Process exited after 3.485 seconds with return value 0
Press any key to continue . . .
```

13. Write a program that takes two numbers as input from the user. The program should calculate the square of each number and then add the two squares together. Display the final result. Example formula:

$$Result = a^2 + b^2$$

```
#include <iostream>
using namespace
std; int main() {
float a, b, result;
  cout << "Enter the first number (a): ";
cin >> a;
  cout << "Enter the second number (b): ";
cin >> b;
```

```
   result = (a * a) + (b * b);

   cout << "The sum of the squares (" << a << "^2 + " << b << "^2) is: " << result <<
endl;

}
```

```
Enter the first number (a): 4
Enter the second number (b): 5
The sum of the squares (4^2 + 5^2) is: 41

--------------------------------
Process exited after 9.376 seconds with return value 0
Press any key to continue . . .
```

14. Write a program that asks the user to enter the price of an item. The program should calculate the total bill amount after adding 10% tax. Display the final bill amount including tax.

$$Total = Price + (Price \times 0.10)$$

```
#include <iostream>

using namespace std;

int main() {

   double price, tax, total;

   cout << "Enter the price of the item: ";

cin >> price;     tax = price * 0.10;     total

= price + tax;

   cout << "Item Price:  " << price << endl;

cout << "Tax (10%):   " << tax << endl;

cout << "Total Bill:  " << total << endl;

}
```

```
Enter the price of the item: 15
Item Price:  15
Tax (10%):   1.5
Total Bill:  16.5

--------------------------------
Process exited after 2.35 seconds with return value 0
Press any key to continue . . .
```

15. Write a program that takes the base and height of a triangle as input from the user. The program should calculate the area of the triangle using the formula. Display the calculated area of the triangle.

$$Area = \frac{1}{2} \times base \times height$$

```
#include <iostream> using namespace std; int main() {

double base, height, area;     cout << "Enter the base of

the triangle: ";     cin >> base;     cout << "Enter the

height of the triangle: ";     cin >> height;     area = 0.5 *

base * height;     cout << "Base:   " << base << endl;

cout << "Height: " << height << endl;     cout << "The

Area of the triangle is: " << area << endl;

}
```

```
Enter the base of the triangle: 15
Enter the height of the triangle: 12
Base:    15
Height: 12
The Area of the triangle is: 90

--------------------------------
Process exited after 10.93 seconds with return value 0
Press any key to continue . . .
```

{end of Lab}